Local ordinary discriminants of Rogers Szegő polynomials at odd roots of unity

An all order derivation with separated and simple overlap hypotheses

Working mathematical note 11 October 2026

1 Purpose and scope

We derive the first moving-root terms of Rogers–Szegő polynomials at a fixed odd root of unity, for every positive block multiplicity A. The finite q-binomial theorem gives an exact diagonal-operator identity. A block-divisibility lemma then controls every order of the formal remainder. In the case of one simple shared root, only one additional product derivative is needed; its coefficient reduces to the classical value at $z = -1$.

Under the stated simplicity and separation hypotheses, these initial forms determine the exact cyclotomic order and the complete leading coefficient of the ordinary discriminant. We retain all signs and distinguish the parameters $q - \omega$, $q^d - 1$, and $\Phi_d(q)$.

The all-order proof underlying this note received an independent scoped acceptance on 11 October 2026. This note reorganizes that accepted proof and its qualifications into a single mathematical account. Historical publication priority remains unresolved. The conclusion is an explicit all-order derivation from classical inputs; it does not assert that the displayed formulas were previously published or establish a worldwide originality claim. Source identities and the finite-control record appear in the appendices.

2 Definitions and conventions

Let K be a characteristic-zero field containing a primitive dth root of unity ω, where $d \geq 3$ is odd. Fix integers $A \geq 1$ and $0 \leq b < d$, and put $n = Ad + b$. Algebraic closures and formal power-series extensions of K may be used to discuss individual roots.

The Gaussian binomial coefficient is the polynomial in $\mathbb{Z}[q]$ represented, for $0 \leq k \leq n$, by

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{j=1}^{k} \frac{1 - q^{n-k+j}}{1 - q^j}.$$

Specialization at a root of unity means evaluation of this polynomial, rather than substitution into an unreduced quotient. Define

$$H_n(z; q) = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k, \qquad D_n(q) = \operatorname{Disc}_z H_n(z; q),$$

$$F(z) = 1 + z^d, \qquad G(z) = H_b(z;\omega), \qquad P_A = \prod_{j=1}^{A} j^j.$$

Every H_n is monic in z. For a monic polynomial $f(z) = \prod_i (z - r_i)$, the ordinary discriminant and the resultant with a monic polynomial g are

$$\mathrm{Disc}(f) = \prod_{i<j} (r_i - r_j)^2, \qquad \mathrm{Res}(f,g) = \prod_i g(r_i).$$

We use $\mathrm{Disc}(1) = \mathrm{Disc}(\text{monic linear}) = 1$ and empty products equal to 1. All discriminants below use the ordinary z derivative, not a q-difference operator.

The monic probabilists' Hermite polynomial is

$$\mathrm{He}_m(y) = m! \sum_{j=0}^{\lfloor m/2 \rfloor} \frac{(-1)^j y^{m-2j}}{2^j j!(m-2j)!}.$$

In particular $\mathrm{He}_0 = 1$, $\mathrm{He}_1 = y$, and $\mathrm{Disc}(\mathrm{He}_A) = P_A$. Write $(a;q)_m = \prod_{j=0}^{m-1}(1 - aq^j)$.

We use formal local parameters

$$q = \omega e^h, \qquad q(\varepsilon) = \omega(1+\varepsilon^2)^{1/d}, \qquad h = \frac{\log(1+\varepsilon^2)}{d}.$$

The binomial and logarithmic series have their constant terms fixed as written. Thus $q(0) = \omega$, $q(\varepsilon)^d - 1 = \varepsilon^2$, and $h = \varepsilon^2/d + O(\varepsilon^4)$. A remainder $O(\varepsilon^s)$ in a local polynomial identity is divisible by ε^s in $K[x][[\varepsilon]]$, after adjoining square roots if convenient. These square roots cancel from the displayed initial forms coefficientwise.

3 The two local theorems

3 1 Separated roots

Theorem 1. For every root α of F, the following coefficient identity holds:

$$H_n(\alpha + \varepsilon x; q(\varepsilon)) = \varepsilon^A (-1)^A d^{A/2} G(\alpha)\, \mathrm{He}_A\left(\frac{\sqrt{d}\, x}{\alpha}\right) + O(\varepsilon^{A+1}). \tag{1}$$

If $G(\alpha) = 0$, the written coefficient vanishes and (1) does not identify the first nonzero term. For the separated discriminant conclusion assume

$$\mathrm{Disc}(G)\,\mathrm{Res}(F,G) \neq 0. \tag{2}$$

Put $M_{\mathrm{sep}} = dA(A-1)/2$. Then

$$D_n(q(\varepsilon)) = \varepsilon^{2M_{\mathrm{sep}}}\big(C_{\mathrm{sep}} + O(\varepsilon^2)\big), \tag{3}$$

$$C_{\mathrm{sep}} = (-1)^{A(d-1)/2} d^{dA(A+1)/2} P_A^d\, \mathrm{Disc}(G)\,\mathrm{Res}(F,G)^{2A}. \tag{4}$$

The coefficient is nonzero, and $\operatorname{ord}_{\Phi_d} D_n = M_{\mathrm{sep}}$. The same separated proof also covers the classical endpoint $d = 1$, $b = 0$; this endpoint is not presented as a new result.

3 2 One simple shared root

Suppose b is odd, so $1 \le b < d$, and write

$$G(z) = (z+1)U(z), \qquad g = U(-1) \neq 0.$$

Set

$$m = \frac{d+b}{2} < d, \qquad W = \frac{(\omega;\omega)_b}{(\omega^2;\omega^2)_m}, \qquad \tau = \frac{dW}{g}, \tag{5}$$

$$J_{A+1}(y;\tau) = \mathrm{He}_{A+1}(y) + A(1-\tau)\,\mathrm{He}_{A-1}(y). \tag{6}$$

Every denominator factor in W is nonzero: since d is odd, ω^2 also has order d, and $m < d$.

Theorem 2. Under $g \neq 0$, the shared-root coefficient identity is

$$H_n(-1+\varepsilon x; q(\varepsilon)) = \varepsilon^{A+1} d^{(A-1)/2} g J_{A+1}(\sqrt{d}\,x;\tau) + O(\varepsilon^{A+2}). \tag{7}$$

Both Hermite summands have parity $A+1$. Therefore the leading factor in (7) is equivalently $(-1)^{A+1} d^{(A-1)/2} g J_{A+1}(-\sqrt{d}\,x;\tau)$, as in the accepted source statement.

For the global overlap discriminant additionally assume

$$\operatorname{Disc}(U)\operatorname{Res}(F,U)\operatorname{Disc}(J_{A+1}) \neq 0. \tag{8}$$

Here $\operatorname{Disc}(J_{A+1})$ means the discriminant in y at the specified value of τ. These hypotheses give exactly one shared residual root, namely -1, simple in G, with distinct first displacements. In fact $\operatorname{Res}(F,U) \neq 0$ already implies $g \neq 0$.

Put $M_{\mathrm{ov}} = dA(A-1)/2 + A$. Then

$$D_n(q(\varepsilon)) = \varepsilon^{2M_{\mathrm{ov}}}(C_{\mathrm{ov}} + O(\varepsilon^2)), \tag{9}$$

$$\begin{aligned} C_{\mathrm{ov}} = {} & (-1)^{A(d-1)/2} d^{dA(A+1)/2+A} P_A^{d-1} \operatorname{Disc}(J_{A+1}) \\ & \cdot \operatorname{Disc}(U) \operatorname{Res}(F,U)^{2A} g^2. \end{aligned} \tag{10}$$

The coefficient is nonzero, and $\operatorname{ord}_{\Phi_d} D_n = M_{\mathrm{ov}}$. No assertion removes (2) or (8), extends this proof to even d, or gives a positive-characteristic unit conclusion.

4 The exact operator and the all order bound

The classical finite q-binomial identity is

$$\mathcal{P}_n(z;q) = \prod_{j=0}^{n-1}(1+q^j z) = \sum_{k=0}^{n} q^{k(k-1)/2} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \tag{11}$$

It is recorded, for example, in Gorodetsky [1, equation (1.5)]. Define diagonal and differential operators on polynomials by

$$T_\omega(z^k) = \omega^{k(k-1)/2} z^k, \qquad \mathcal{L} = z^2 \partial_z^2,$$

and set $R_n(z,h) = T_\omega^{-1} \mathcal{P}_n(z; \omega e^h)$. Comparing coefficients in (11) gives the exact formal identity

$$H_n(z; \omega e^h) = \exp(-h\mathcal{L}/2) R_n(z,h). \tag{12}$$

This takes place in the h-adic module of polynomials in z of degree at most n. The coefficient of a fixed h power uses only finitely many terms of the exponential.

Oddness of d is essential to the next step. The multiplier ratio for z^{k+d} and z^k is

$$\omega^{(k+d)(k+d-1)/2 - k(k-1)/2} = \omega^{dk + d(d-1)/2} = 1.$$

Consequently T_ω and its inverse commute with multiplication by z^d and by every power of F. For even d the same ratio is -1.

Lemma 3. If $R_n(z,h) = \sum_{r \geq 0} h^r R_{n,r}(z)$, then

$$F^{\max(A-r,0)} \mid R_{n,r} \quad \text{for every } r \geq 0, \qquad R_{n,0} = F^A G. \tag{13}$$

Proof. Partition the first Ad factors of $\mathcal{P}_n$ into A consecutive blocks of length d, followed by a residual block of length b. At $h = 0$, each full block equals F. In a term of the rth product derivative, at most r full blocks are differentiated. At least $A - r$ intact blocks remain if $r \leq A$. Apply T_ω^{-1}, which commutes with their product. Division by $r!$ does not affect divisibility. The residual finite q-binomial identity gives $R_{n,0} = F^A G$. This proves (13) at every order. □

At a root α of F, put $u = z - \alpha$ and assign u weight 1 and h weight 2. The rth summand of R_n has weighted order at least

$$2r + \max(A - r, 0). \tag{14}$$

This is at least $A + 1$ for $r \geq 1$ and at least $A + 2$ for $r \geq 2$, including $r > A$.

In these local coordinates,

$$h\mathcal{L} = h\alpha^2 \partial_u^2 + 2h\alpha u \partial_u^2 + h u^2 \partial_u^2. \tag{15}$$

The three terms preserve weight, increase it by one, and increase it by two, respectively. Hence $\exp(-h\mathcal{L}/2)$ preserves every weighted remainder ideal. On a fixed leading weight its action is $\exp(-h\alpha^2 \partial_u^2/2)$. Its weight-preserving part is a finite sum on each weighted homogeneous polynomial, because repeated second derivatives terminate. The full expression is also well defined h-adically by (12). No analytic convergence is used.

5 Proof of the initial forms

5 1 The separated coefficient

Since $F'(\alpha) = -d/\alpha$, the weight-A part of R_n is

$$(-d/\alpha)^A G(\alpha) u^A.$$

All positive-r parts have larger weight by (14). The finite polynomial identity

$$\exp(-s\partial_u^2/2)u^A = \sum_{j=0}^{\lfloor A/2\rfloor} \frac{A!(-s)^j u^{A-2j}}{2^j j!(A-2j)!} \tag{16}$$

equals $s^{A/2}\,\mathrm{He}_A(u/\sqrt{s})$. Apply (16) with $s = h\alpha^2$, then put $u = \varepsilon x$ and $h = \varepsilon^2/d + O(\varepsilon^4)$. The result is (1). Formula (16) also explains directly why square roots cancel.

5 2 The first product derivative at the shared root

Let $\mathcal{D} = z\partial_z$ and define

$$E_b(z) = \prod_{r=0}^{b-1}(1+\omega^r z),$$

$$S(z) = \sum_{r=0}^{d-1} r\omega^r z \prod_{\substack{0\le s<d\\ s\neq r}} (1+\omega^s z),$$

with S_b defined by the same sum and product over $0 \le r, s < b$. The derivative of full block i, indexed from $i = 0$, is $S + id^2 z^d$. Therefore

$$\begin{aligned}\partial_h \mathcal{P}_n(z;\omega e^h)\big|_{h=0} = F^{A-1}\left(AE_bS + \frac{d^2A(A-1)}{2}z^dE_b\right)\\ + F^A\left(Ad\mathcal{D}E_b + S_b\right).\end{aligned} \tag{17}$$

Apply T_ω^{-1}. The z^dE_b term acquires a factor G, which vanishes at -1. The final F^A term has weight at least $A+2$ after multiplication by h. Thus, with $u = z+1$ and

$$c_b = (T_\omega^{-1}(E_bS))(-1),$$

the only weight-$(A+1)$ contribution from $hR_{n,1}$ is $Ad^{A-1}c_bhu^{A-1}$. This statement also holds for $A = 1$; the term with factor $A(A-1)$ is then zero.

We can determine c_b using $A = 1$. Differentiating (12) gives

$$R_{d+b,1} = \partial_h H_{d+b}(z;\omega e^h)\big|_{h=0} + \frac{\mathcal{L}}{2}(FG).$$

Because $FG = dg\,u^2 + O(u^3)$, the second term equals dg at $z = -1$. The classical Gaussian value is

$$H_{2m}(-1;q) = (q;q^2)_m = \frac{(q;q)_{d+b}}{(q^2;q^2)_m}, \qquad 2m = d+b. \tag{18}$$

Odd-degree values vanish. These identities also follow from the three-term recurrence displayed in Appendix A; see Schultz–Walker [2, printed page 2]. The numerator on the right of (18) contains exactly one vanishing factor at $q = \omega$, namely $1-q^d$. The remaining factors multiply to $d(\omega;\omega)_b$, since $\prod_{j=1}^{d-1}(1-\omega^j) = d$. Also $\partial_h(1-e^{dh})|_{h=0} = -d$. Therefore

$$H_{d+b}(-1;\omega e^h) = -d^2 W h + O(h^2), \qquad c_b = dg - d^2 W = dg(1-\tau). \tag{19}$$

The weight-$(A+1)$ static term is $d^A g u^{A+1}$, while every $r \geq 2$ term has weight at least $A+2$. The complete initial form of R_n is consequently

$$d^A g \left(u^{A+1} + A(1-\tau) h u^{A-1} \right). \tag{20}$$

At $\alpha = -1$, apply $\exp(-h\partial_u^2/2)$ to (20) and substitute $u = \varepsilon x$, $h = \varepsilon^2/d + O(\varepsilon^4)$. Formula (16) gives (7). Both the factor d in $\tau = dW/g$ and the factor $1/d$ in the leading term of h are necessary.

6 Ordinary discriminants and signs

Under the respective guards, each simple root of the initial polynomial lifts by the formal implicit-function theorem to a root branch in an algebraic closure with coefficients in formal ε series. In the separated case there are A branches at each of the d roots of F, together with b residual branches. In the overlap case there are $A+1$ branches at -1, A at each other root of F, and $b-1$ residual branches. These counts account for all n roots.

6 1 Separated clusters

If $h_1, \ldots, h_A$ are the roots of He_A, the first displacements at α are $\alpha h_j/\sqrt{d}$. The product of squared differences within all clusters is therefore, to leading order,

$$\varepsilon^{dA(A-1)} d^{-dA(A-1)/2} P_A^d. \tag{21}$$

The omitted α powers cancel: $\prod_{F(\alpha)=0} \alpha = -1$ and $A(A-1)$ is even. The cross-cluster product is $\mathrm{Disc}(F)^{A^2}$, where

$$\mathrm{Disc}(F) = (-1)^{(d-1)/2} d^d. \tag{22}$$

Residual pairs and pairs between clusters and residual roots contribute $\mathrm{Disc}(G)\,\mathrm{Res}(F,G)^{2A}$. Multiplying these terms, using $A^2 \equiv A \pmod 2$, proves (3) and (4).

6 2 The shared cluster

The first displacements in (7) are $j_k/\sqrt{d}$, where the j_k are the roots of J_{A+1}. The parity convention with negative argument gives the same set. Put $F_0 = F/(z+1)$. Then

$$F_0(-1) = d, \qquad \mathrm{Disc}(F_0) = (-1)^{(d-1)/2} d^{d-2}.$$

The product of the roots of F_0 is 1. The within-cluster contribution is

$$\varepsilon^{2M_{\mathrm{ov}}} d^{-M_{\mathrm{ov}}} P_A^{d-1}\,\mathrm{Disc}(J_{A+1}). \tag{23}$$

Cross-cluster pairs contribute

$$\mathrm{Disc}(F_0)^{A^2} d^{2A(A+1)} = (-1)^{A(d-1)/2} d^{dA^2+2A}. \tag{24}$$

The residual contribution is $\mathrm{Disc}(U)\,\mathrm{Res}(F,U)^{2A}g^2$. The extra factor g^2 accounts for the extra root in the shared cluster. Combining (23) and (24) proves (9) and (10).

In both cases, nonvanishing of the displayed coefficient establishes the exact order. If a guard fails, these root-lifting arguments do not certify that order. Although individual root branches need not be even in ε, the discriminant is even because $q(\varepsilon)$ is even. This gives the normalized remainders $O(\varepsilon^2)$ in (3) and (9).

7 Conversion between local parameters

Let (M, C) mean either $(M_{\mathrm{sep}}, C_{\mathrm{sep}})$ or $(M_{\mathrm{ov}}, C_{\mathrm{ov}})$, under its hypotheses, and put $t = q^d - 1$. The formulas above say

$$D_n(q) = t^M(C + O(t)) \quad \text{locally at } q = \omega. \tag{25}$$

Since $t = d\omega^{d-1}(q-\omega) + O((q-\omega)^2)$, the coefficient of $(q-\omega)^M$ is

$$(d\omega^{d-1})^M C. \tag{26}$$

It is generally different from C. Let $Q_d(q) = (q^d - 1)/\Phi_d(q)$. The integral polynomial $D_n(q)$ has the same exact root multiplicity at every primitive dth root by Galois conjugacy. Monic division gives

$$D_n(q) = \Phi_d(q)^M E(q), \qquad E(q) \in \mathbb{Z}[q], \qquad E(\omega) = Q_d(\omega)^M C. \tag{27}$$

These assertions are in characteristic zero. Nonvanishing of C there does not imply that its reduction is a unit in any specified positive characteristic.

8 Classical antecedents and logical dependence

The input (11) is classical. The static identity $H_n(z;\omega) = F(z)^A G(z)$ is the usual root-of-unity factorization; (12)–(13) also derive it directly. The special value (18), the Hermite discriminant, and the root-product discriminant rule are established ingredients. The proof here supplies the block filtration and the first surviving coefficients explicitly.

Schultz–Walker [2, Theorem 1.1, printed page 3] provides a relevant all-order vanishing antecedent. If $\alpha^d = -1$ has order $2c$, then $c \mid d$ and d/c is odd. In their theorem take $k = d/c$, their auxiliary exponent and congruence-order parameter equal to zero, derivative index r, and weight polynomial $P(s) = s(s-1)\cdots(s-\ell+1)$. Up to the nonzero factor $\alpha^{-\ell}$, their sum is $\partial_q^r \partial_z^\ell H_n(\alpha; q)$. Their bound becomes $n \geq (\ell + 2r + 1)d$. Hence

$$\partial_q^r \partial_z^\ell H_n(\alpha;\omega) = 0 \quad \text{whenever } 2r + \ell < A. \tag{28}$$

Their primitive-choice passage is on printed page 13. The stated implication covers roots of smaller order as well as primitive $2d$th roots. The cited theorem gives vanishing; it does not specify the surviving coefficient.

The logical distinctions matter. For $b = 0$ and $A \geq 2$, adding $\lambda(q^d - 1)F^{A-2}$ to a monic polynomial family preserves its static specialization and all below-weight-A vanishings, but changes its weight-A initial form. At the shared root, adding $\lambda(q^d - 1)F^{A-1}$ similarly changes weight $A + 1$ while preserving

the lower vanishings. These are logical counterexamples to determining a coefficient from vanishing data alone. The perturbed families are not asserted to be Rogers–Szegő polynomials.

Conversely, a full local initial-form identity implies its lower-weight mixed-derivative vanishings. The coordinate change $q - \omega = \omega h + O(h^2)$ is invertible and filtered; substitution $u = \varepsilon x$, $h = \varepsilon^2/d + O(\varepsilon^4)$ separates the coefficients of each weight by powers of x. This local converse does not recover the full Schultz–Walker theorem. Likewise, the guarded initial forms imply the discriminant coefficients, but a single discriminant coefficient does not determine all initial polynomials or imply the operator identity.

The accepted implication is therefore

$$\begin{aligned} &\text{finite product identity and proved block filtration} \\ &\quad + \text{shared root special value when needed} \\ &\qquad \Longrightarrow \text{local initial forms} \\ &\quad \Longrightarrow \text{guarded discriminant formulas.} \end{aligned}$$

This derivation settles the previously unworked all-A implication route. It does not settle when the resulting formulas were first published. The finite controls in Appendix A support the audit and cannot establish the all-A claim by themselves.

9 Remaining comparisons and extensions

Historical comparison still requires the exact six-page body of Zhedanov's 1997 paper [4], the printed page 360 image of Lubinsky–Saff [3], and substantive bodies along the relevant Lubinsky–Saff citation chain. Missing or inaccessible text supplies no evidence that a theorem is absent. The comparison of the complex-parameter disk and inverse-pair applications is a separate open priority question.

The local theorem itself retains several boundaries. Multiple common residual roots, a repeated root at -1, or a repeated root of J_{A+1} would require a different initial-form analysis. The proof does not remove the composite residual squarefreeness or resultant guards. Even d changes the periodicity used in (13). Uniform estimates when degree grows, and arithmetic conclusions requiring positive-characteristic units, need additional arguments. None of these extensions is claimed here.

For $A = 1$, the separated discriminant is already the static product-discriminant consequence; the simple-overlap calculation reduces to (18) and a Taylor expansion. For arbitrary A, the accepted result is a fully written all-order coefficient derivation under the guards. Any assessment of its publication contribution should separate correctness, explicit derivability, historical priority, and the value of particular applications.

References

[1] Ofir Gorodetsky. *q-congruences, with applications to supercongruences and the cyclic sieving phenomenon.* arXiv:1805.01254v2, equation (1.5), preprint page 2. The citation records the classical finite q-binomial identity and does not assign its origin to this paper. Primary preprint

[2] Andrew Schultz and Robert Walker. *A generalization of the Gaussian formula and a q-analog of Fleck's congruence.* arXiv:1202.0199v1, printed pages 2–3 and 13. Published in *Journal of Number Theory* 133(11) (2013), 3717–3738, DOI 10.1016/j.jnt.2013.05.003. Page locators here refer to the preprint. Primary preprint

[3] D. S. Lubinsky and E. B. Saff. *Convergence of Padé approximants of partial theta functions and the Rogers–Szegő polynomials* (1987). Historical comparison remains incomplete at the source scope stated

in Section 9. Publisher record

[4] A. Zhedanov. *Gauss sums and orthogonal polynomials* (1997). The exact body remains an outstanding comparison source. Publisher record

Appendix A Finite controls and reproducibility

The independent review used a freshly written three-term recurrence implementation:

$$H_0 = 1, \quad H_1 = 1 + z, \quad H_n = (1 + z)H_{n-1} + (q^{n-1} - 1)zH_{n-2}.$$

It computed exact rational truncated ε series in $\mathbb{Q}[w]/(w^d - 1)$, then reduced modulo $\Phi_d(w)$. It did not generate the tested jets using the operator proof or the author's q-Pascal program. The author's code was read as inert text during that review, not imported or executed.

All 16 saved moving-root controls passed. In the table, s specifies the tested root $\alpha = -\omega^s$. "Other root" means evaluation of formula (1); it does not claim that every residual guard holds globally for that parameter triple.

A	d	b	s	Local formula	Initial weight
1	3	1	0	Shared root	2
2	3	2	2	Other root	2
4	3	1	0	Shared root	5
6	3	1	0	Shared root	7
10	3	1	0	Shared root	11
12	3	0	2	Other root	12
5	5	1	0	Shared root	6
4	5	3	0	Shared root	5
3	5	4	1	Other root	3
3	7	5	0	Shared root	4
4	7	2	3	Other root	4
3	9	7	0	Shared root	4
4	9	2	3	Other root	4
2	15	3	5	Other root	2
1	15	13	0	Shared root	2
7	1	0	0	Classical endpoint	7

Five direct ordinary-discriminant computations also passed the exact cyclotomic valuation and both coefficient conversions:

(A, d, b)	Degree n	Exact Φ_d order
$(1, 3, 2)$	5	0
$(1, 5, 1)$	6	1
$(2, 3, 2)$	8	3
$(2, 3, 1)$	7	5
$(1, 5, 3)$	8	1

Six mutation controls detected the intended errors: omitting the heat term; replacing τ by 1; omitting d from τ; assuming the same periodicity for even d; omitting the $q-\omega$ scaling; and dropping the separated resultant guard. Constant-polynomial discriminants were explicitly set to 1.

This appendix reports the sealed independent-control run. The full exact outputs, executable source, and run log are retained in the independent review archive identified below. They test bounded examples; the proof for every A is Lemma 3 and Sections 5–7.

Appendix B Source identities and review boundary

The original proof archive and independent review archive retain their existing identities. This working note is a separate document, not a replacement or a retrospective edit of either archive. The prior independent acceptance concerns the exact proof bytes below; editorial consistency of this note must be distinguished from a new independent mathematical audit.

All order implication archive

F020_All_Order_Implication_Provisional.zip

Library identity: libfile_f90f326d1b0c8191a838d4cb2e7fab76, version 0.

SHA256: `79273ecedccd68bf0d174c843eb5cecc9d32cbb596b115519d26da11352b9ffd`

Accepted proof member ALL_ORDER_IMPLICATION.md

SHA256: `8ca570c7e7c41c50827b1dea3feb931cbcfe0fdab3c3a2016c6e7d0989126abc`

Independent review archive

F020_All_Order_Independent_Review.zip

Library identity: libfile_770d307623188191a6408f4915be0687, version 0.

SHA256: `45671b26ad19d741ca9a8b08da11db23d946a60f6758f18b98544b2a1a8f860e`

Independent review member INDEPENDENT_REVIEW.md

SHA256: `aa6fab3492f265fca7a667a213f57f1453dcfb280d94280ce74b330b353bffd3`

Independent control output INDEPENDENT_CONTROLS.json

SHA256: `778d7585963a6bef193280895da256cc6cac6ee6c6b5ab95e8a2f8ac0216f0ab`

Independent control source independent_controls.py

SHA256: `2ddd2cbef7c652152aae23f7877494bec320ff6c4e8ec4e9c56cd2847cca14a0`

The independent verdict accepts the characteristic-zero all-A implication in the separated and one-simple-overlap scope, with no correction required to its displayed formulas. It checked selected primary PDF text for [1] and [2]. Its attempted primary page-image retrieval failed; it does not claim a rendered-page inspection or an exhaustive whole-paper negative result.

For the earlier separated audit, the recovered file has the following SHA256.

`f093175f18d1941a99a605a5d816b1db31e5b2c4161ee7df4bf29291ff8eff3c`

The catalogue lists a distinct export SHA256.

`4b86b3078bbdf644a3317e67449a974c356e8b2dd8a9f66a393df71341d3e8d9`

It is not silently identified with those bytes. The original separated and overlap reports, and their independent acceptance records, remain unchanged.